

GALAXY CLUSTERS AND SUPERCLUSTERS

MOST GALAXIES ARE FOUND IN CLUSTERS OF ANYTHING FROM A FEW DOZEN TO A THOUSAND OR MORE GALAXIES GROUPED TOGETHER BY GRAVITY. CLUSTERS BLEND AT THE EDGES TO FORM SUPERCLUSTERS, CREATING WEBLIKE FILAMENTS AROUND APPARENTLY EMPTY SPACE.

Galaxy clusters are the largest structures in the Universe that are created by the force of gravity, pulling galaxies together over millions of light-years of space to form huge concentrations of mass and matter. Because of this, they tend to fill a fairly similar volume of space (10–20 million light-years across) regardless of how many galaxies they contain. Superclusters and even bigger structures reflect the large-scale distribution of matter caused by the Big Bang itself (see pp.70–71).

Types of clusters

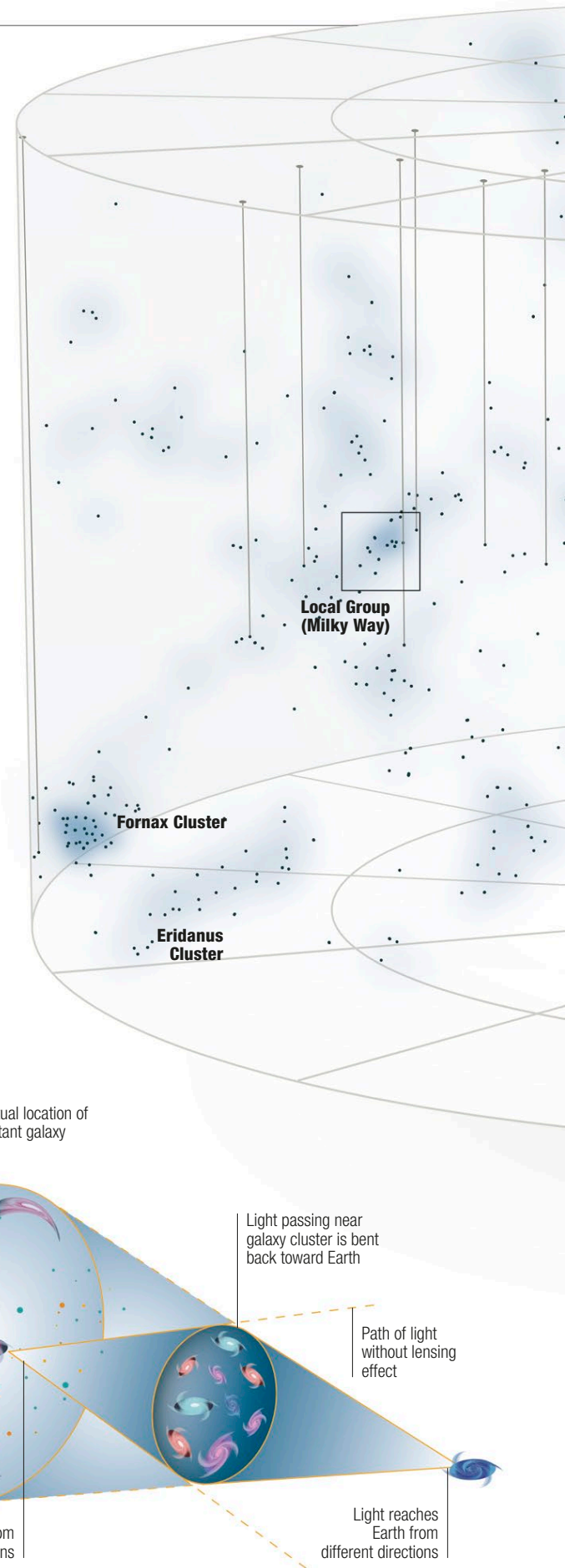
The Milky Way Galaxy belongs to a low-density cluster called the Local Group—it is one of three large spirals surrounded by 50 or so smaller galaxies, most of which are tiny and faint dwarf systems. Most small clusters seem to follow this pattern and are dominated by spirals and irregular galaxies. However, clusters containing large numbers of galaxies are very different, tending to be dominated by elliptical galaxies full of red and yellow stars. This is an important clue that ellipticals are created by the collision and merger of other types of galaxies within dense clusters.



◀ **Coma cluster**
Roughly 320 million light-years from Earth, the Coma Cluster is a group of about 1,000 galaxies, most of which are elliptical or lenticular. This infrared image reveals large numbers of dwarf galaxies, too faint to detect in visible light.



◀ **Lensing effect**
The distorted images caused by lensing allow astronomers to use gravity as a natural telescope for spotting faint and distant objects. They can also be used to work out how mass is distributed within the lensing cluster.



△ **Gravitational lensing**
The huge concentration of mass in galaxy clusters gives rise to an effect known as gravitational lensing. Large masses alter the shape of space near to them (see p.73). Light passing through a massive galaxy cluster changes direction, so galaxies beyond the cluster look distorted and sometimes magnified.

